Graph Neural Networks for Misinformation Propagation Analysis

Riley Demoson, Avery Testwell
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Abstract

Misinformation spreads through social networks along user connections. We model propagation graphs with graph neural networks and detect misinformation from the spreading structure rather than the text.

Keywords: misinformation propagation, graph neural network, social networks

1. Introduction

Text-based detectors ignore how misinformation travels. We study graph neural networks on propagation graphs of shared posts.

2. Related Work

Hybrid text models such as Hybrid BERT-CNN Architecture for Misinformation Detection reach high accuracy but use no propagation structure. Surveys such as A Survey of Fake News Detection Using Machine Learning mention graph methods only briefly.

3. Methodology

Each story is a propagation graph of users and reshares. A graph convolutional network produces a graph embedding that is classified as true or false.

4. Results

The GCN reached 91.7 percent accuracy using structure alone. Combining it with text features raised accuracy to 94.2 percent.

5. Limitations

Propagation data is often unavailable for new stories. Real-time evaluation is limited.

6. Conclusion

Propagation structure carries a strong signal for misinformation detection.

7. Future Work

Future work should combine text and graph models in a hybrid system and test on live streams.

References

[1] A Survey of Fake News Detection Using Machine Learning, Journal of Demonstration Studies, 2021.
[2] Hybrid BERT-CNN Architecture for Misinformation Detection, Journal of Demonstration Studies, 2022.

Note: this is a SYNTHETIC demonstration paper with fictional authors and invented results, created to demo ResearchGraph. It is not a real publication.